

We mentioned in the introduction to Chapter 3 that one reason for using algebraic polynomials to approximate an arbitrary set of data is that, given any continuous function defined on a closed interval, there exists a polynomial that is arbitrarily close to the function at every point in the interval. Also, the derivatives and integrals of polynomials are easily obtained and evaluated. It should not be surprising, then, that many procedures for approximating derivatives and integrals use the polynomials that approximate the function.

Numerical Differentiation (Sayisal Türev)

The derivative of the function f at x_0 is

$$f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}.$$

This formula gives an obvious way to generate an approximation to $f'(x_0)$; simply compute

$$\frac{f(x_0 + h) - f(x_0)}{h}$$

for small values of h . Although this may be obvious, it is not very successful, due to our old nemesis round-off error. But it is certainly a place to start.

As noted above, high-accuracy divided-difference formulas can be generated by including additional terms from the Taylor series expansion. For example, the forward Taylor series expansion can be written as

$$\underline{f(x_{i+1})} = \underline{f(x_i)} + \underline{f'(x_i)h} + \frac{f''(x_i)}{2!}h^2 + \dots$$

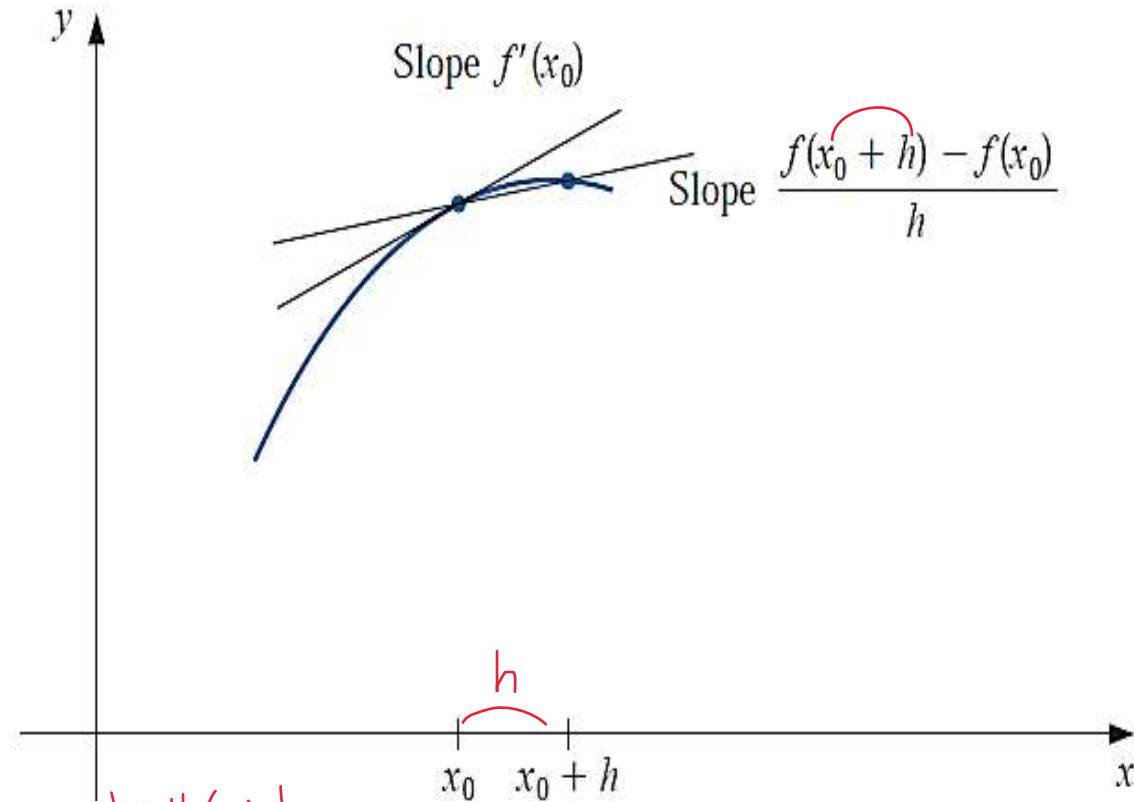
which can be solved for

$$\underline{f'(x_i)} = \frac{f(x_{i+1}) - f(x_i)}{h} - \frac{f''(x_i)}{2}h + O(h^2)$$

When x is x_0 ,

$$f'(x_0) = \frac{f(x_0 + h) - f(x_0)}{h} - \frac{h}{2} f''(\xi)$$

$$x_0 < \xi < x_0 + h \quad \max |f''(\xi)| = M$$



For small values of h , the difference quotient $[f(x_0 + h) - f(x_0)]/h$ can be used to approximate $f'(x_0)$ with an error bounded by $M|h|/2$, where M is a bound on $|f''(x)|$ for x between x_0 and $x_0 + h$. This formula is known as the **forward-difference formula** if $h > 0$ and the **backward-difference formula** if $h < 0$. $h > 0 \rightarrow$ forw

Example 1 Use the forward-difference formula to approximate the derivative of $f(x) = \ln x$ at $x_0 = 1.8$ using $h = 0.1$, $h = 0.05$, and $h = 0.01$, and determine bounds for the approximation errors.

Solution The forward-difference formula

ilen for k

$$\frac{f(1.8 + h) - f(1.8)}{h}$$

$$\frac{f(x_0 + h) - f(x_0)}{h} = f'(x_0)$$

$$f(1.9) = \ln(1.9)$$

with $h = 0.1$ gives

$$f'(1.8) = \frac{\ln 1.9 - \ln 1.8}{0.1} = \frac{0.64185389 - 0.58778667}{0.1} = 0.5406722.$$

$$f'(x) = \frac{1}{x} = x^{-1}$$

Because $f''(x) = -1/x^2$ and $1.8 < \xi < 1.9$, a bound for this approximation error is

$\frac{1}{x^2}$ $\frac{1}{\xi^2}$ $\frac{|h f''(\xi)|}{2} = \frac{|h|}{2\xi^2} < \frac{0.1}{2(1.8)^2} = 0.0154321$ error bound $\rightarrow M$

$\xi_t = \frac{|0.55 - 0.54|}{|0.55|}$ relative error

$f'(1.8) = 0.55$

The approximation and error bounds when $h = 0.05$ and $h = 0.01$ are found in a similar manner and the results are shown in Table 4.1.

$$0.55 - 0.54 = 0.01 < M$$

h	$f(1.8 + h)$	$\frac{f(1.8 + h) - f(1.8)}{h}$	$\frac{ h }{2(1.8)^2}$
0.1	0.64185389	0.5406722	0.0154321
0.05	0.61518564	0.5479795	0.0077160
0.01	0.59332685	0.5540180	0.0015432

Since $f'(x) = 1/x$, the exact value of $f'(1.8)$ is 0.555, and in this case the error bounds are quite close to the true approximation error.

To obtain general derivative approximation formulas, suppose that $\{x_0, x_1, \dots, x_n\}$ are $(n+1)$ distinct numbers in some interval I and that $f \in C^{n+1}(I)$.

$$f(x) = \sum_{k=0}^n f(x_k) L_k(x) + \frac{(x-x_0) \cdots (x-x_n)}{(n+1)!} f^{(n+1)}(\xi(x)),$$

for some $\xi(x)$ in I , where $L_k(x)$ denotes the k th Lagrange coefficient polynomial for f at $x_0, x_1, \dots, x_n$. Differentiating this expression gives

$$f'(x) = \sum_{k=0}^n f(x_k) L'_k(x) + D_x \left[\frac{(x-x_0) \cdots (x-x_n)}{(n+1)!} \right] f^{(n+1)}(\xi(x)) + \frac{(x-x_0) \cdots (x-x_n)}{(n+1)!} D_x [f^{(n+1)}(\xi(x))].$$

We again have a problem estimating the truncation error unless x is one of the numbers x_j . In this case, the term multiplying $D_x[f^{(n+1)}(\xi(x))]$ is 0, and the formula becomes

$$f'(x_j) = \sum_{k=0}^n f(x_k) L'_k(x_j) + \frac{f^{(n+1)}(\xi(x_j))}{(n+1)!} \prod_{\substack{k=0 \\ k \neq j}}^n (x_j - x_k),$$

which is called an $(n+1)$ -point formula to approximate $f'(x_j)$.

$(n+1)$ -point formula

$$f(x) \approx p_2(x) = L_0 \cdot f_0 + L_1 \cdot f_1 + L_2 \cdot f_2 \quad f'(x) = L'_0 f_0 + L'_1 f_1 + L'_2 f_2$$

We first derive some useful three-point formulas and consider aspects of their errors.

Because

$$L_0(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)}, \quad \text{we have} \quad L'_0(x) = \frac{2x - x_1 - x_2}{(x_0-x_1)(x_0-x_2)}.$$

Similarly,

$$L'_1(x) = \frac{2x - x_0 - x_2}{(x_1-x_0)(x_1-x_2)} \quad \text{and} \quad L'_2(x) = \frac{2x - x_0 - x_1}{(x_2-x_0)(x_2-x_1)}.$$

Hence, from Eq. (4.2),

$$f'(x_j) = f(x_0) \left[\frac{2x_j - x_1 - x_2}{(x_0-x_1)(x_0-x_2)} \right] + f(x_1) \left[\frac{2x_j - x_0 - x_2}{(x_1-x_0)(x_1-x_2)} \right] + f(x_2) \left[\frac{2x_j - x_0 - x_1}{(x_2-x_0)(x_2-x_1)} \right] + \frac{1}{6} f^{(3)}(\xi_j) \prod_{\substack{k=0 \\ k \neq j}}^2 (x_j - x_k), \quad (4.3)$$

for each $j = 0, 1, 2$, where the notation ξ_j indicates that this point depends on x_j .

Three-Point Formulas (Üç Nokta Formülleri)

The formulas from Eq. (4.3) become especially useful if the nodes are equally spaced, that is, when

$$x_0, \quad \underline{x_1 = x_0 + h} \quad \text{and} \quad \underline{x_2 = x_0 + 2h}, \quad \text{for some } h \neq 0.$$

We will assume equally-spaced nodes throughout the remainder of this section.

Using Eq. (4.3) with $x_j = x_0, x_1 = x_0 + h$, and $x_2 = x_0 + 2h$ gives

$$f'(x_0) = \frac{1}{h} \left[-\frac{3}{2} f(x_0) + 2f(x_1) - \frac{1}{2} f(x_2) \right] + \frac{h^2}{3} f^{(3)}(\xi_0).$$

Doing the same for $x_j = x_1$ gives

$$f'(x_1) = \frac{1}{h} \left[-\frac{1}{2} f(x_0) + \frac{1}{2} f(x_2) \right] - \frac{h^2}{6} f^{(3)}(\xi_1),$$

and for $x_j = x_2$,

$$f'(x_2) = \frac{1}{h} \left[\frac{1}{2} f(x_0) - 2f(x_1) + \frac{3}{2} f(x_2) \right] + \frac{h^2}{3} f^{(3)}(\xi_2).$$

Since $x_1 = x_0 + h$ and $x_2 = x_0 + 2h$, these formulas can also be expressed as

$$f'(x_0) = \frac{1}{h} \left[-\frac{3}{2} f(x_0) + 2f(x_0 + h) - \frac{1}{2} f(x_0 + 2h) \right] + \frac{h^2}{3} f^{(3)}(\xi_0),$$

$$f'(x_0 + h) = \frac{1}{h} \left[-\frac{1}{2} f(x_0) + \frac{1}{2} f(x_0 + 2h) \right] - \frac{h^2}{6} f^{(3)}(\xi_1),$$

and

$$f'(x_0 + 2h) = \frac{1}{h} \left[\frac{1}{2} f(x_0) - 2f(x_0 + h) + \frac{3}{2} f(x_0 + 2h) \right] + \frac{h^2}{3} f^{(3)}(\xi_2).$$

As a matter of convenience, the variable substitution x_0 for $x_0 + h$ is used in the middle equation to change this formula to an approximation for $f'(x_0)$. A similar change, x_0 for $x_0 + 2h$, is used in the last equation. This gives three formulas for approximating $f'(x_0)$:

$$f'(x_0) = \frac{1}{2h}[-3f(x_0) + 4f(x_0 + h) - f(x_0 + 2h)] + \frac{h^2}{3}f^{(3)}(\xi_0),$$

$$f'(x_0) = \frac{1}{2h}[-f(x_0 - h) + f(x_0 + h)] - \frac{h^2}{6}f^{(3)}(\xi_1),$$

and

$$f'(x_0) = \frac{1}{2h}[f(x_0 - 2h) - 4f(x_0 - h) + 3f(x_0)] + \frac{h^2}{3}f^{(3)}(\xi_2).$$

Finally, note that the last of these equations can be obtained from the first by simply replacing h with $-h$, so there are actually only two formulas:

Üç Nokta Uç Formülü

Three-Point Endpoint Formula

$$\bullet f'(x_0) = \frac{1}{2h}[-3f(x_0) + 4f(x_0 + h) - f(x_0 + 2h)] + \frac{h^2}{3}f^{(3)}(\xi_0), \quad (*) \quad (4.4)$$

where ξ_0 lies between x_0 and $x_0 + 2h$.

Üç Nokta Orta Formülü

Three-Point Midpoint Formula

$$\bullet f'(x_0) = \frac{1}{2h}[f(x_0 + h) - f(x_0 - h)] - \frac{h^2}{6}f^{(3)}(\xi_1), \quad (*) \quad (4.5)$$

where ξ_1 lies between $x_0 - h$ and $x_0 + h$.

Five-Point Formulas (Bes Nokta Formülleri)

The methods presented in Eqs. (4.4) and (4.5) are called **three-point formulas** (even though the third point $f(x_0)$ does not appear in Eq. (4.5)). Similarly, there are **five-point formulas** that involve evaluating the function at two additional points. The error term for these formulas is $O(h^4)$. One common five-point formula is used to determine approximations for the derivative at the midpoint.

Five-Point Midpoint Formula

- $$f'(x_0) = \frac{1}{12h} [f(x_0 - 2h) - 8f(x_0 - h) + 8f(x_0 + h) - f(x_0 + 2h)] + \frac{h^4}{30} f^{(5)}(\xi), \quad (4.6)$$

where ξ lies between $x_0 - 2h$ and $x_0 + 2h$.

Five-Point Endpoint Formula

- $$f'(x_0) = \frac{1}{12h} [-25f(x_0) + 48f(x_0 + h) - 36f(x_0 + 2h) + 16f(x_0 + 3h) - 3f(x_0 + 4h)] + \frac{h^4}{5} f^{(5)}(\xi), \quad (4.7)$$

where ξ lies between x_0 and $x_0 + 4h$.

Example 2 Values for $f(x) = xe^x$ are given in Table 4.2. Use all the applicable three-point and five-point formulas to approximate $f'(2.0)$.

Table 4.2

x	$f(x)$
1.8	10.889365
1.9	12.703199
2.0	14.778112
2.1	17.148957
2.2	19.855030

Solution The data in the table permit us to find four different three-point approximations. We can use the endpoint formula (4.4) with $h = 0.1$ or with $h = -0.1$, and we can use the midpoint formula (4.5) with $h = 0.1$ or with $h = 0.2$.

Using the endpoint formula (4.4) with $h = 0.1$ gives

$$\frac{1}{0.2}[-3f(2.0) + 4f(2.1) - f(2.2)] = 5[-3(14.778112) + 4(17.148957) - 19.855030] = 22.032310,$$

$\swarrow$
 $2h$

and with $h = -0.1$ gives 22.054525.

Using the midpoint formula (4.5) with $h = 0.1$ gives

$$\frac{1}{0.2}[f(2.1) - f(1.9)] = 5(17.148957 - 12.703199) = 22.228790,$$

and with $h = 0.2$ gives 22.414163. $\frac{1}{0.4} [f(2.2) - f(1.8)]$

The only five-point formula for which the table gives sufficient data is the midpoint formula (4.6) with $h = 0.1$. This gives

$$\frac{1}{1.2} [f(1.8) - 8f(1.9) + 8f(2.1) - f(2.2)] = \frac{1}{1.2} [10.889365 - 8(12.703199) + 8(17.148957) - 19.855030] = 22.166999$$

$\downarrow$ $2-2h$ $\downarrow$ $2-h$ $\downarrow$ $2+h$ $\downarrow$ $2+2h$

$\downarrow$ $12 \cdot h$

$f'(2)$

If we had no other information we would accept the five-point midpoint approximation using $h = 0.1$ as the most accurate, and expect the true value to be between that approximation and the three-point mid-point approximation that is in the interval $[22.166, 22.229]$.

The true value in this case is $f'(2.0) = (2 + 1)e^2 = 22.167168$, so the approximation errors are actually:

Three-point endpoint with $h = 0.1$: 1.35×10^{-1} ;

Three-point endpoint with $h = -0.1$: 1.13×10^{-1} ;

Three-point midpoint with $h = 0.1$: -6.16×10^{-2} ;

Three-point midpoint with $h = 0.2$: -2.47×10^{-1} ;

Five-point midpoint with $h = 0.1$: 1.69×10^{-4} .

$$f(x) = xe^x$$

$$f'(x) = e^x + xe^x = e^x(1+x)$$

$$\hookrightarrow 22.167168 - 22.166999 = f'(2)$$



Methods can also be derived to find approximations to higher derivatives of a function using only tabulated values of the function at various points. The derivation is algebraically tedious, however, so only a representative procedure will be presented.

Expand a function f in a third Taylor polynomial about a point x_0 and evaluate at $x_0 + h$ and $x_0 - h$. Then

$$f(x_0 + h) = f(x_0) + f'(x_0)h + \frac{1}{2}f''(x_0)h^2 + \frac{1}{6}f'''(x_0)h^3 + \frac{1}{24}f^{(4)}(\xi_1)h^4$$

and

$$f(x_0 - h) = f(x_0) - f'(x_0)h + \frac{1}{2}f''(x_0)h^2 - \frac{1}{6}f'''(x_0)h^3 + \frac{1}{24}f^{(4)}(\xi_{-1})h^4,$$

where $x_0 - h < \xi_{-1} < x_0 < \xi_1 < x_0 + h$.

If we add these equations, the terms involving $f'(x_0)$ and $-f'(x_0)$ cancel, so

$$f(x_0 + h) + f(x_0 - h) = 2f(x_0) + f''(x_0)h^2 + \frac{1}{24}[f^{(4)}(\xi_1) + f^{(4)}(\xi_{-1})]h^4.$$

Solving this equation for $f''(x_0)$ gives

$$f''(x_0) = \frac{1}{h^2}[f(x_0 - h) - 2f(x_0) + f(x_0 + h)] - \frac{h^2}{24}[f^{(4)}(\xi_1) + f^{(4)}(\xi_{-1})]. \quad (4.8)$$

Suppose $f^{(4)}$ is continuous on $[x_0 - h, x_0 + h]$. Since $\frac{1}{2}[f^{(4)}(\xi_1) + f^{(4)}(\xi_{-1})]$ is between $f^{(4)}(\xi_1)$ and $f^{(4)}(\xi_{-1})$, the Intermediate Value Theorem implies that a number ξ exists between ξ_1 and ξ_{-1} , and hence in $(x_0 - h, x_0 + h)$, with

$$f^{(4)}(\xi) = \frac{1}{2}[f^{(4)}(\xi_1) + f^{(4)}(\xi_{-1})].$$

Second Derivative Midpoint Formula (İkinci Türev Orta Formülü)

- $$f''(x_0) = \frac{1}{h^2}[f(x_0 - h) - 2f(x_0) + f(x_0 + h)] - \frac{h^2}{12}f^{(4)}(\xi), \quad (4.9)$$

for some ξ , where $x_0 - h < \xi < x_0 + h$.

If $f^{(4)}$ is continuous on $[x_0 - h, x_0 + h]$ it is also bounded, and the approximation is $O(h^2)$.

Example 3 In Example 2 we used the data shown in Table 4.3 to approximate the first derivative of $f(x) = xe^x$ at $x = 2.0$. Use the second derivative formula (4.9) to approximate $f''(2.0)$.

Solution The data permits us to determine two approximations for $f''(2.0)$. Using (4.9) with $h = 0.1$ gives

$$\begin{aligned} \frac{1}{0.01}[f(1.9) - 2f(2.0) + f(2.1)] &= 100[12.703199 - 2(14.778112) + 17.148957] \\ &= 29.593200, \end{aligned}$$

and using (4.9) with $h = 0.2$ gives

$$\begin{aligned} \frac{1}{0.04}[f(1.8) - 2f(2.0) + f(2.2)] &= 25[10.889365 - 2(14.778112) + 19.855030] \\ &= 29.704275. \end{aligned}$$

Because $f''(x) = (x + 2)e^x$, the exact value is $f''(2.0) = 29.556224$. Hence the actual errors are -3.70×10^{-2} and -1.48×10^{-1} , respectively. ■

x	$f(x)$
1.8	10.889365
1.9	12.703199
2.0	14.778112
2.1	17.148957
2.2	19.855030